

Instructions for running code

To run the main code `uv run assignment_2.py`

1. Sketches of inverted pendulum

Included at the end of document

2. Visualization of the regions of attraction

At the end of report

3. Choice of Poincare section

The poincare section I am using is when θ is 0. This is a good poincare section because $\theta = 0$ is parallel to the paths and the paths are generally similarly spaced. $\theta = 0$ is also an easy place to analyze and far from the impacts so they won't have a significant effect. Using a poincare section also reduces the problem to a single state because the only free variable is angular velocity.

4. Grid Resolution Varification

I created a lookup table that uses the nearest grid state to choose each step. The grid has to be fine enough that the rounding consistently gives accurate results but runs quickly. I picked 50 random starting points to test each grid resolution and simulated to see how many steps they took to reach a standstill and if they took the expected amount of steps. I used a 80% threshold for both categories because some of the values near the edge of the first node could not reach node 0.

		velocity spacing (rad/s)	alpha spacing (rad)	reaches RoA	steps as pre- dicted	passes	steps from 3.0(pre- dicted / actual)
n_velocity	n_alpha						
120	60	0.037	0.00095	65.2%	65.2%	no	3 / inf
180	60	0.025	0.00095	67.4%	67.4%	no	3 / 3
240	60	0.019	0.00095	84.8%	84.8%	yes	3 / 3
360	60	0.012	0.00095	84.8%	84.8%	yes	3 / 3
480	60	0.009	0.00095	91.3%	91.3%	yes	3 / 3

		velocity spacing (rad/s)	alpha spacing (rad)	reaches RoA	steps as pre- dicted	passes	steps from 3.0(pre- dicted / actual)
n_velocity	n_alpha						
240	20	0.019	0.00295	80.4%	76.1%	no	3 / 3
240	40	0.019	0.00144	71.7%	71.7%	no	3 / 3
240	50	0.019	0.00114	84.8%	82.6%	yes	3 / 3
240	60	0.019	0.00095	84.8%	84.8%	yes	3 / 3

I decided to use $n_velocity = 240$ and $n_alpha = 50$ for my grid. This grid value reached my desired threshold and every finer grid i tested also passed.

5. Plot the trajectory for max and min steps

At the end of report

6. Visulaization of how many steps

At the end of report